

UNIT – 1
ASSESSMENT – 2
SOLUTIONS

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1. Introduction

The **Backtracking Search Algorithm** is a depth-first search technique used to solve **Constraint Satisfaction Problems (CSPs)**.

It assigns values one by one and checks whether the current assignment satisfies all constraints.

- If the assignment is valid, the algorithm continues.
- If a constraint is violated, it **backtracks** and tries another assignment.

This continues until a valid solution is found.

Problem Formulation

Initial State

No doctor is assigned to any shift.

D1 = ?

D2 = ?

D3 = ?

Variables

- D1
- D2
- D3

Domain

Each doctor can be assigned one of the following shifts:

- Morning
- Afternoon
- Night

Constraints

- $D1 \neq \text{Night}$
- $D3 \neq \text{Morning}$
- D2 must work before D3
- Only one doctor per shift
- Every doctor gets exactly one shift

Backtracking Process

Step 1

Assign

$D1 \rightarrow \text{Morning}$

Constraint Check

- D1 cannot work Night ✓
- Morning available ✓

Current Assignment

$D1 = \text{Morning}$

Step 2

Assign

D2 → Afternoon

Constraint Check

- Shift available ✓
- D2 before D3 ✓

Current Assignment

D1 = Morning

D2 = Afternoon

Step 3

Assign

D3 → Night

Constraint Check

- D3 cannot work Morning ✓
- D2 before D3 ✓
- Night available ✓

Current Assignment

D1 = Morning

D2 = Afternoon

D3 = Night

All constraints satisfied.

No backtracking is required because the first assignment itself satisfies all conditions.

2. Introduction

The **Breadth-First Search (BFS)** algorithm is an uninformed search algorithm that explores nodes **level by level**. It always finds the **shortest path** in an unweighted graph (or when every move has the same cost).

Since every movement costs **1**, BFS guarantees the shortest path.

Although the question mentions the **Manhattan Distance heuristic**, BFS **does not use heuristics** while searching. The Manhattan Distance can be used only to estimate the distance from a node to the goal.

Grid Representation

Example 5×5 Grid

```
+---+---+---+---+---+
| S | | X | | |
+---+---+---+---+
| | | X | | |
+---+---+---+---+
| | | X | | |
+---+---+---+---+
| X | X | | |
+---+---+---+---+
| | | | X | G |
```

+---+---+---+---+---+

Where:

- **S** = Start
- **G** = Goal
- **X** = Obstacle

State Space

Each cell in the grid represents a state.

Example:

(0,0), (0,1), (0,2) ...

Initial State

Robot Position = S (0,0)

Goal State

Robot reaches G (4,4)

Operators

The robot can move

- Up
- Down
- Left
- Right

provided the cell is inside the grid and is not an obstacle.

Step-by-Step BFS Traversal

Step 1

Queue

$[(0,0)]$

Expand

$(0,0)$

Add

$(0,1)$

$(1,0)$

Queue

$[(0,1),(1,0)]$

Step 2

Expand

$(0,1)$

New node

$(1,1)$

Queue

$[(1,0),(1,1)]$

Step 3

Expand

(1,0)

New node

(2,0)

Queue

[(1,1),(2,0)]

Step 4

Expand

(1,1)

New node

(2,1)

Queue

[(2,0),(2,1)]

Step 5

Expand

(2,0)

Obstacle encountered on one side.

Continue.

Queue

[(2,1)]

Step 6

Continue expanding until Goal is reached.

Eventually,
Goal Found

3. Introduction

An autonomous rescue robot is an **Intelligent Agent** that senses its environment, makes decisions, and performs actions to achieve a goal.

Since the environment is **unknown and changes dynamically**, the robot must continuously gather information and update its path while searching for survivors.

The **Online Search Agent** combined with **Uniform Cost Search (UCS)** is the most suitable approach.

(a) Constraint Analysis

Constraint 1

Movement only in four directions

Effect:

The robot can only move:

- Up
- Down
- Left
- Right

Diagonal movement is not allowed.

Constraint 2

Blocked cells (Obstacles)

Effect:

The robot cannot pass through blocked rooms.

It must search for an alternative path.

Constraint 3

Limited sensing capability

Effect:

The robot only knows the status of adjacent rooms.

It cannot see the entire building.

Therefore, it must continuously explore.

Constraint 4

Movement cost = 1

Effect:

Every movement increases the total path cost.

The robot should avoid unnecessary movement.

Constraint 5

Risky zones cost = +2

Effect:

Risky areas are allowed but expensive.

The robot should avoid them unless they produce a cheaper overall route.

Constraint 6

Online Search

Effect:

The robot continuously updates its map while moving.

New obstacles can be discovered at any time.

The robot replans whenever necessary.

(b) Solution Approach

Search Strategy

Uniform Cost Search (UCS) with Online Search Agent

Working

Step 1

Start from position S

↓

Step 2

Observe adjacent cells

↓

Step 3

Calculate movement cost

↓

Step 4

Ignore blocked cells

↓

Step 5

Choose the path having the lowest cumulative cost



Step 6

Update knowledge after every movement



Step 7

Repeat until Goal **G** is reached

Example Search

S



Move Right



Observe New Cell



Move Down



Avoid Obstacle



Move Right



Enter Safe Cell



Goal G

The robot always selects the path with the minimum total cost.

(c) AI Concepts

Intelligent Agent

The rescue robot is an Intelligent Agent because it:

- Perceives the environment using sensors.
- Makes decisions independently.
- Performs actions.
- Learns new information while moving.

Rationality

The robot is rational because it always chooses the action that minimizes:

- Travel cost
- Risk
- Time

while maximizing the chance of reaching the survivor safely.

Environment Type

Property	Type
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Observability	Partially Observable
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Agents	Single Agent
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Environment	Dynamic
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State Space	Sequential
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Knowledge	Unknown
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Actions	Deterministic
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(d) Justification

The **Online Search Agent with Uniform Cost Search** is the most suitable because:

- The robot does not know the complete environment.
- New obstacles may appear.

- The robot continuously updates its knowledge.
- UCS guarantees the least-cost path.
- The robot avoids risky zones whenever possible.
- Suitable for real disaster rescue operations.

Advantages

- Finds minimum-cost path.
- Handles unknown environments.
- Adapts dynamically.
- Safe navigation.
- Efficient rescue planning.

Conclusion

The autonomous rescue robot successfully applies Artificial Intelligence concepts to navigate safely in a disaster environment.

The **Online Search Agent** enables dynamic decision-making, while **Uniform Cost Search** ensures that the robot reaches the survivor with the minimum possible cost.